The Price of a Summary:

Information-Theoretic Limits of Agent Oversight

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Abstract

A human overseeing a fleet of software agents reads digests, not transcripts. We show that this practice obeys exact information-theoretic limits, and we price them. Four results: (i) any digest that *guarantees* the overseer catches all k critical artifacts among N while opening only m must carry at least $\log_2 \binom{N}{k} - \log_2 \binom{m}{k}$ bits — a floor that survived a pre-registered falsification experiment (0/16 violations, including an oracle encoder); (ii) for strict total reader orders, one scalar head serves both if and only if their orders coincide — and the overseer and the successor agent provably are not, with the joint floor *super*-additive ($\approx 2.13\times$, not $2\times$); (iii) the folklore product stakes × irreversibility × anomaly is expected loss of passing when anomaly is a calibrated posterior; the inspect-or-pass decision also prices attention and false alarms; (iv) a two-constraint rate–distortion program $R(\delta, f)$ — false-negative rate priced separately from flag rate — has a closed form for Bernoulli sources, with the covering floor as its zero-miss corner, and a new zoom-advantage theorem: adaptive halving identifies the k criticals inside F flagged items in at most $2k\lceil \log_2(F/k) \rceil + 4k$ group opens, a bound whose leading constant is achieved. Every number is tagged as externally recomputable or as regenerating from a named script at a fixed seed.

1 Introduction

The scene. An operator supervises forty coding agents overnight. By morning the fleet has produced hundreds of artifacts — branches, migration scripts, config edits, deleted test suites. Two of them are critical: if the operator does not open exactly those two before the morning merge, an irreversible mistake ships. The operator will read one screen of digest and open perhaps eight items. The question this paper answers with theorems rather than taste: *what must that screen carry, what can it never do, and what does each relaxation buy?*

Express lane. One sentence per result, formal statements in the boxes. A zero-miss digest is a message that must single out a covering m-subset, so it costs $\log_2 \binom{N}{k} - \log_2 \binom{m}{k}$ bits (box in §2); one scalar head serves two strict-order readers iff their orders coincide, and oversight's two readers are not (box in §3); the inspect-or-pass rule is a likelihood-ratio test, making "anomaly" a calibration obligation (box in §4); and pricing misses and flags separately yields a closed-form frontier $R(\delta, f)$ whose flagged set is then searched in $2k\lceil \log_2(F/k) \rceil + 4k$ adaptive opens — the one theorem newly proved in this paper (boxes in §5).

The idea in one breath: a summary is a channel, an overseer is a decoder with an open budget, and every oversight guarantee therefore has an exact price in bits, opens, and misses — prices we compute, and one of which we bound with a new, constant-explicit group-testing theorem.

Contributions.

1. **The information floor, falsified-against** (§2): the covering bound $B^\star = \log_2 \binom{N}{k} - \log_2 \binom{m}{k}$, plus a pre-registered experiment in which no encoder — oracle, noisy, or random

— beat it (0/16 violations at N=60, k=2, m=8 [internal, `a7_experiment.py`]). The meter must charge the literal message carrying critical-set identity; side-channel scores invalidate the comparison.

2. **The split-digest characterization** (§3): a scalar head serves two strict-order readers iff their induced orders coincide; the operator (regret-shaped) and successor (fit-shaped) readers order constructible pairs oppositely, so no shared head exists — and the joint zero-miss floor is super-additive: $\mathrm{Floor}(N, 2k, m) \approx 2.13 \times \mathrm{Floor}(N, k, m)$, a ratio-over-double of 1.067 at N=60, k=2, m=8 that exceeds 1 whenever $2k \le m < N$ and can grow without bound along varying N, m, k (1.224 at k=4, m=8; 1.392 at k=10, m=20) [verified].
3. **The derived regret head** (§4): stakes × irreversibility × anomaly is exactly expected unrecoverable loss iff anomaly is the calibrated posterior $\Pr(\texttt{bad} \mid \text{evidence})$; the optimal surfacing rule is a likelihood-ratio threshold, and raising the miss-to-false-alarm cost ratio *lowers* the bar to inspect.
4. **The two-constraint frontier and the zoom theorem** (§5): a custom rate–distortion program $R(\delta, f)$ with false negatives priced separately from flag rate, solved in closed form for Bernoulli sources (the optimizing joint is fully pinned); and a new worst-case bound for the second stage — adaptive halving finds all k positives among F flagged items in at most $2k\lceil \log_2(F/k) \rceil + 4k$ group opens versus F flat, proved by a charging argument, shown tight in its leading constant, and consistent with the measured 15.3× advantage at $(F, k) = (2500, 10)$ [internal, `b1_frontier.py`].

Results 1–3 were derived and verified in the program's execution reports; result 4's frontier was computed there but its zoom bound existed only as a simulation regularity — the theorem and proof in §5.2 are new to this paper. Throughout, numbers are tagged **[verified]** (externally recomputable from the closed form) or **[internal]** (regenerates from the named script at seed 20260816).

2 The information floor, and its falsification test

The analogy, mapped by relations. A digest is not a description of the night's work; it is a *message* that selects, from a codebook shared with the overseer, which m items to open (Figure 1). The relation that transfers: a B-bit message distinguishes at most 2^B codewords, and each codeword — an m-subset — covers only $\binom{m}{k}$ of the $\binom{N}{k}$ ways the k critical items could be placed. The predictable misread, preempted: the floor does *not* say a digest needs $B^\star$ bits to be useful — it prices the *zero-miss guarantee* only; cheaper digests are fine the moment misses are tolerated (§5 prices exactly that).

Theorem 1 (information floor). Fix N items containing an arbitrary unknown k-subset T of critical items, and an open budget m. Any digest scheme — an encoder e mapping the corpus to B bits and a data-independent decoder d mapping B bits to an m-subset to open — that guarantees $T \subseteq d(e(\cdot))$ for *every* placement of T must have

$$B \;\ge\; B^\star(N, k, m) \;=\; \log_2 \binom{N}{k} - \log_2 \binom{m}{k}.$$

Proof. The decoder realizes at most 2^B distinct m-subsets; an m-subset contains $\binom{m}{k}$ of the $\binom{N}{k}$ possible k-subsets; zero miss requires the realized m-subsets to jointly cover all placements, so $2^B \binom{m}{k} \ge \binom{N}{k}$. □

Numbers by hand. At N=60, k=2, m=8: $\log_2 \binom{60}{2} = \log_2 1770 = 10.79$, $\log_2 \binom{8}{2} = \log_2 28 = 4.81$, so $B^\star = 5.98$ bits [verified] (Table 1 collects this and every other headline

Quantity (N=60, k=2, m=8)	Value	Provenance
Floor $B^\star$	5.98 bits	[verified]
Floor, disjoint readers ($2k$=4; §3)	12.77 bits	[verified]
Ratio $\text{Floor}_{2k}/\text{Floor}_k$	2.13×	[verified]
Super-additivity $\text{Floor}_{2k}/(2\,\text{Floor}_k)$	1.067	[verified]
Oracle violations of the floor	0/16	[internal, `a7_experiment.py`]

Table 1: Every headline number in §2–§3, with its evidence tier. The four rows above the inner rule are closed-form quantities recomputable by hand from Theorem 1; the row below it is a finite simulation count. The pairing that matters: the joint floor is 12.77 bits where naive doubling budgets 11.96.

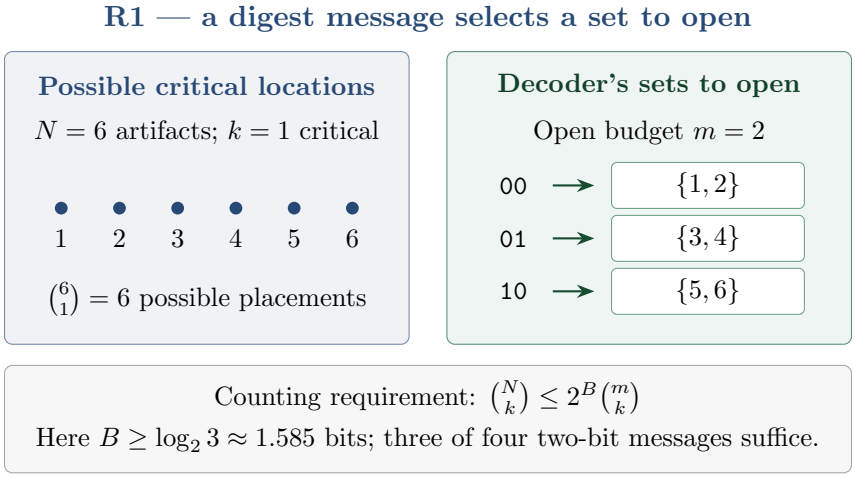


Figure 1: A complete covering in the $N = 6$, $k = 1$, $m = 2$ case. The three displayed messages name disjoint sets and cover every placement. The counting inequality is necessary for every codebook; overlap between sets can prevent a codebook from attaining it in other cases.

number with its evidence tier). Now you try: N=100, k=2, m=10 — compute $\log_2 4950$ and $\log_2 45$ and check you get a floor just under 6.8 bits.

The falsification experiment. Theorem 1 makes a sharp prediction: under uniformly distributed critical placements, a B-bit digest's probability of opening every critical item is *at most* $\min(1, 2^B\binom{m}{k}/\binom{N}{k})$. Codeword cover sets can overlap, so equality need not be attainable; the miss probability is at least one minus this bound. A scheme below that miss floor would refute the formalization. We implemented three encoders — an *oracle* that sees the true set and addresses the best codeword its B bits can reach, a *noisy* encoder (feature SNR ≈ 1), and a *random* one — against a shared randomized-cover codebook, 4000 placements per budget point, sweeping B through the floor at N=60, k=2, m=8. Result: **0/16 violations**; the oracle hugs the floor from above, the others pay a premium [internal, `a7_experiment.py`]. The floor's regime behavior is Figure 2: it falls as the open budget grows and reaches zero only at $m = N$.

Meter boundary. The B bits must select a codeword in the encoder–decoder codebook. Supplying perfect per-item scores outside that message leaks the critical-set identity through an unpriced side channel; charging only for rank ties does not test the floor.

3 When one ranking cannot serve two readers

The analogy, mapped by relations. A scalar summary head is a thermometer: one number, one order, every reader who consults it ranked the same way (Figure 3). That is fine when the

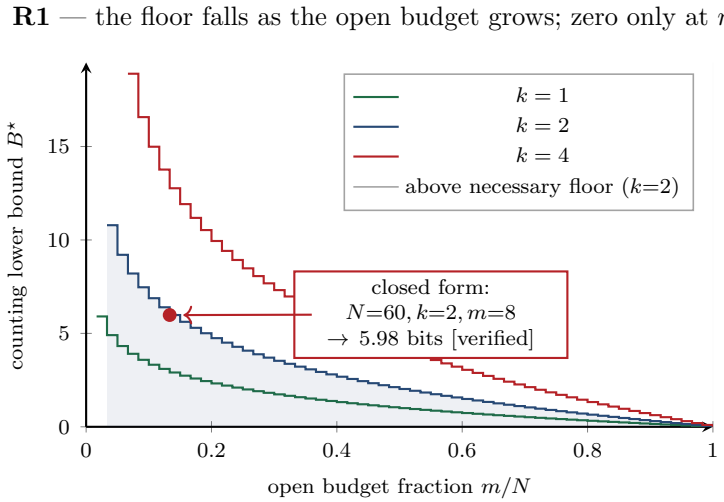


Figure 2: Regime diagram for the necessary counting floor $B^\star$ against the open-budget fraction m/N (here N=60, k=2; the marked point is the experiment's m=8, $B^\star$=5.98 bits [verified]). The lower bound is steep where budgets are tight and vanishes when the overseer opens everything. Points above the bound are not automatically achievable: codeword cover sets can overlap.

readers want the same thing — a nurse and a pharmacist both want the hottest patient first. It fails the moment two readers want quantities that do not move together. Every compaction here is read twice: by the *successor* — the next agent instance, which wants what is worth continuing (fit-shaped, forgone continuation value) — and by the *operator*, who wants what is dangerous to ignore (risk-shaped, regret-if-ignored). The relation that transfers is not "a summary resembles a temperature" but the harder fact underneath it: a scalar serving two readers with strict total orders must induce their common order; ties in a reader's preferences may be refined without conflict. Using one shared selection order for both readers assumes that their strict preferences never cross. A shared store or a jointly encoded summary does not make that assumption. The misread to preempt: this is not a claim that scalars are too coarse — a scalar with *infinite* precision fails identically, because the obstruction is ordinal, not numeric.

Theorem 2 (comonotone characterization). Let X be a finite set of candidate items. A scalar head $h : X \to \mathbb{R}$ with one shared deterministic tie rule *serves* a reader with loss L_r if its top-m set realizes r's unique optimal selection at every budget m. For two readers with *strict total orders*, one head serves both iff their orders agree on every pair (the strict-order instance of comonotonicity [12, 13]). This condition can fail for successor and operator objectives: a working-state item can rank above an audit item for continuation, while the audit item ranks above the working state for oversight. If those strict preferences hold, no shared head serves both.

Proof. ($\Leftarrow$) Give items descending scalar values in the common strict order. ($\Rightarrow$) A strict order has a unique top-m set at each budget; these nested sets recover its pairwise order. A shared head yielding both readers' unique top-m sets must therefore reproduce the same order. A pair with those opposite strict preferences therefore rules out a shared head. □

A complete budgeted instance. Let the evidence be $E = \{w, a\}$ and allow only extractive summaries $D \subseteq E$ with $|D| \leq 1$. The working item w contains the next task's required bit. With w the successor predicts that bit exactly; without it the successor predicts uniformly. Its base-two KL loss against the correct point mass is therefore $L_s(D) = 0$ if $w \in D$ and 1 otherwise. An operator who must inspect audit item a incurs the stipulated oversight loss $L_o(D) = 0$

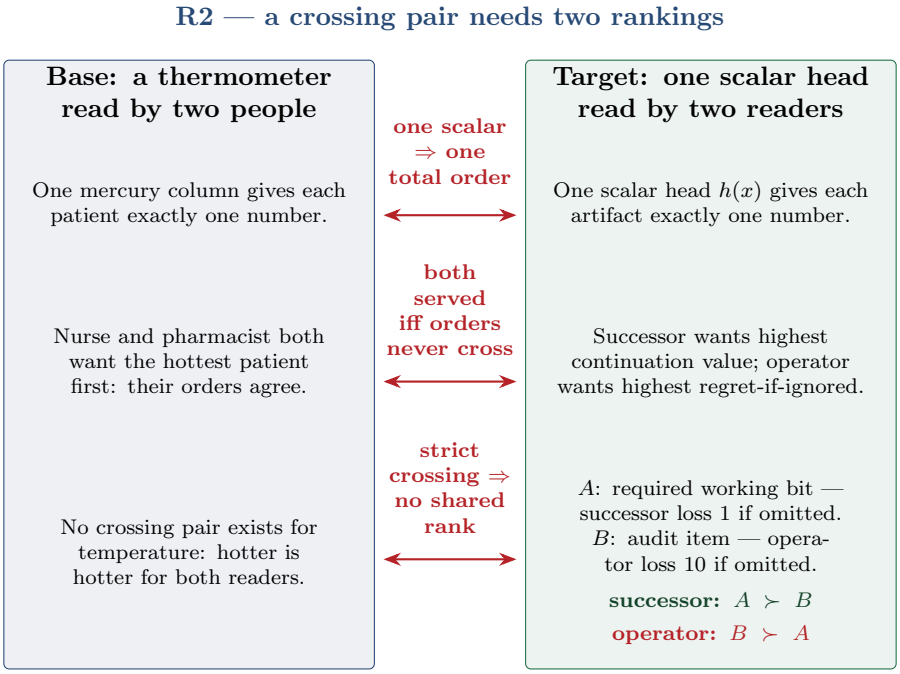


Figure 3: Theorem 2 concerns one shared selection order, with a common tie rule, serving both readers at every budget. A strict crossing forbids such an order. The worked instance allows one extractive item: A avoids a successor prediction loss of one bit; B avoids a stipulated oversight loss of ten. A jointly encoded representation or a larger extraction budget lies outside that counterexample. The relation map does not establish how frequently the conflict occurs in deployed fleets.

if $a \in D$ and 10 otherwise. The three feasible loss pairs are $(1, 10)$ for the empty summary, $(0, 10)$ for $\{w\}$, and $(1, 0)$ for $\{a\}$. No feasible summary minimizes both. The weight 10 is an audit priority, not a claim that an abandoned event has irreversible future effects. With budget two, or an encoding class that fits both items in one message, this counterexample disappears. Thus the summary class, loss functions, and budget are part of the statement.

Claim 1 (Super-additive split penalty)**.** *Let* $\mathrm{Floor}(N, k, m) = B^\star(N, k, m)$ *from Theorem 1. A joint zero-miss digest serving two readers with* disjoint *critical* k*-sets must catch their union and so needs* $\mathrm{Floor}(N, 2k, m)$ *bits, and*

$$\frac{\mathrm{Floor}(N, 2k, m)}{2\,\mathrm{Floor}(N, k, m)} > 1 \quad \textit{for every } N, k, m \textit{ with } 2k \le m < N,$$

because $\mathrm{Floor}(N, t, m) = \sum_{i=0}^{t-1} \log_2((N-i)/(m-i))$ *has strictly increasing summands for* $N > m$*, so its second block of* k *terms exceeds its first. The ratio can grow without bound along families with changing* N, m, k*; fixed* N, m *admit only finitely many* k*. At* $N{=}60, m{=}8$ *it runs* 1.029 *(*$k{=}1$*),* 1.067 *(*$k{=}2$*),* 1.122 *(*$k{=}3$*),* 1.224 *(*$k{=}4$*); at* $N{=}60, m{=}20$ *it reaches* 1.392 *by* $k{=}10$ *[verified]. At the reference regime* $N{=}60, k{=}2, m{=}8$ *(Table 1): joint floor* 12.77 *bits versus* $2 \times 5.98 = 11.96$*; ratio* 2.13×*, super-additivity* 1.067 *[verified]. This compares one zero-miss selection for the union under a shared* m*-open budget with two applications of the single-reader floor. The denominator adds two lower bounds, each allowing* m *opens, hence* $2m$ *opens in total; it is not an attainable coding cost or an equal-total-budget architecture comparison. It does not prove separate models or stores are necessary (Figure 4).*

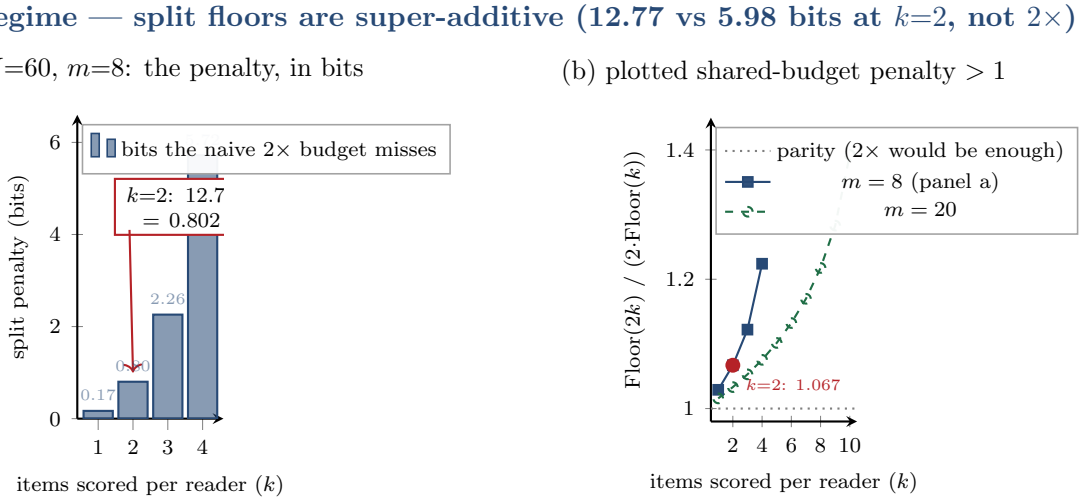


Figure 4: Split-floor comparison under one shared m-open budget. Panel (a): the excess bits for a union of disjoint critical sets over twice the single-set floor, at $N{=}60, m{=}8$ — 0.80 bits at $k{=}2$ and 5.72 bits at $k{=}4$. Panel (b): the ratio over double across the plotted finite regimes. It exceeds 1 at every plotted point, reaching 1.224 at $m{=}8, k{=}4$ and 1.392 at $m{=}20, k{=}10$ [verified]. This does not require separate summary models or stores.

What it buys. Distinct reader-specific rankings are necessary when the strict crossing pair occurs; they may use one vector-valued representation and one durable store. The strict crossing pair rules out a shared head even if other items tie. A strict preference by one reader and a tie by the other are compatible: one head can refine that tie.

4 The regret head is derived, not designed

Which flagged items should actually interrupt a human? Fleet dashboards rank by stakes × irreversibility × anomaly — three tuned knobs. We show the product form is a theorem under one commitment, which converts "anomaly" from a vibe into a measured obligation.

The analogy, mapped by relations. A surfacing rule is a smoke detector (Figure 5). It has one dial, and the right setting of that dial is fixed not by how much smoke there is but by the *ratio* of two costs: the house burning down against burnt toast waking you at 3am. The relations that transfer are three. Expected loss is magnitude times probability, so the alarm must weigh what it would cost to be wrong, not merely how unusual the reading is. Raising the cost of the fire *lowers* the threshold at which the alarm should sound — the dial moves toward hair-trigger, not away from it. And a detector that cries wolf is not miscalibrated in *sensitivity*; it is miscalibrated in *probability*, because the number it reports is not the chance of a fire. That third relation is this section's thesis: once "anomaly" is required to be $\Pr(\mathsf{bad} \mid \text{evidence})$, tuning stops being the job and calibration starts. The misread to preempt: the detector analogy invites the thought that a better sensor removes the trade-off. It does not — a perfect sensor still needs the cost ratio to know where to trip, and the threshold below is a statement about costs, not about evidence quality.

Theorem 3 (regret head). For item x let $C_{\text{miss}}(x) = \text{stakes}(x) \cdot \text{irrev}(x)$ be the unrecoverable loss of passing a bad x, let $a(x) = \Pr(\mathsf{bad} \mid \text{evidence}(x))$ be a *calibrated* posterior, $c_{\text{att}} \geq 0$ the marginal attention cost, and $C_{\text{fa}} \geq 0$ the non-attention cost of inspecting a good item. The expected unrecoverable loss of passing is exactly $C_{\text{miss}}(x)\, a(x)$, and the

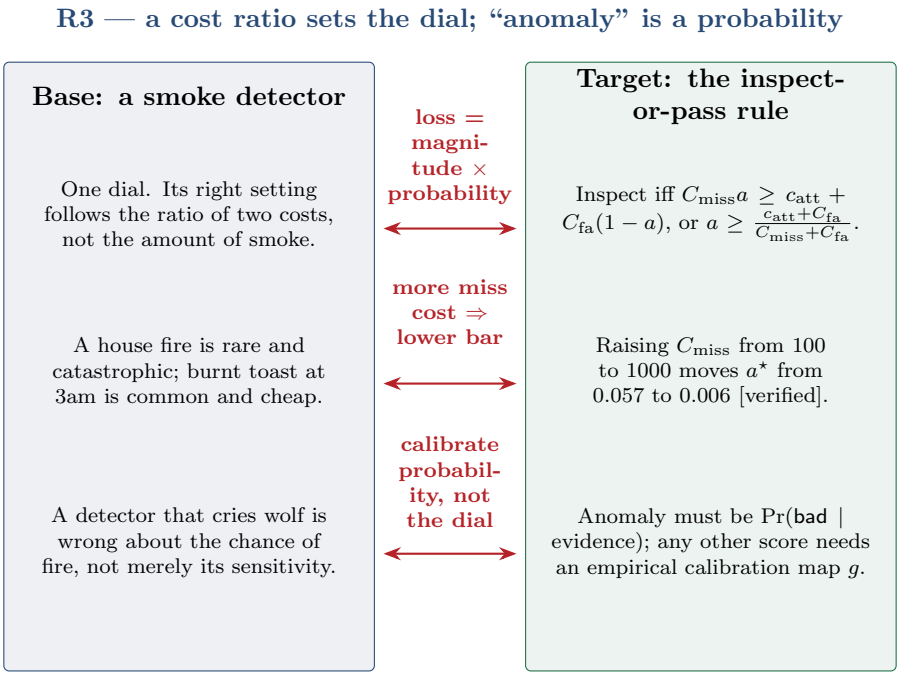


Figure 5: The relation-map for Theorem 3. Three relations transfer, and they are the section in order: expected loss is magnitude times probability, so the rule must weigh what being wrong costs and not merely how unusual the reading is; raising the miss cost *lowers* the threshold at which to act (drawn in row 2, $a^\star$ moving from 0.057 to 0.006 as C_{miss} goes 100 → 1000 [verified]); and a rule that surfaces too much is miscalibrated in probability rather than over-sensitive, which is why anomaly = Pr(bad | evidence) is an obligation and not a naming convention.

Bayes-optimal policy inspects iff

$$C_{\text{miss}}(x)\,a(x) \;\geq\; c_{\text{att}} + C_{\text{fa}}\big(1-a(x)\big), \qquad \text{equivalently} \qquad a(x) \;\geq\; \frac{c_{\text{att}}+C_{\text{fa}}}{C_{\text{miss}}(x)+C_{\text{fa}}},$$

when $C_{\text{miss}} + C_{\text{fa}} > 0$. For $C_{\text{miss}} > c_{\text{att}}$ and $a < 1$, the odds form is $a/(1-a) \geq (c_{\text{att}} + C_{\text{fa}})/(C_{\text{miss}} - c_{\text{att}})$. At $a = 1$, compare C_{miss} directly with c_{att}; when $C_{\text{miss}} = C_{\text{fa}} = 0$, inspection only ties if attention is free. The product stakes × irrev × anomaly is expected loss of passing when anomaly is the calibrated posterior a. With item-dependent costs, inspection priority follows net benefit $(C_{\text{miss}} + C_{\text{fa}})a - (c_{\text{att}} + C_{\text{fa}})$; any other score first needs an empirical calibration map $g(s) = a$.

Proof. Expected loss of passing: $C_{\text{miss}}(x)a(x)$. Expected cost of inspecting: $c_{\text{att}} + C_{\text{fa}}(1 - a(x))$. Compare and collect a terms. The stated odds form follows only when $C_{\text{miss}} > c_{\text{att}}$ and $a < 1$. □

Numbers by hand. C_{miss}=100, c_{att}=1, C_{fa}=5: inspect when $a \geq 6/105 \approx 0.057$ [verified]. Raising C_{miss} to 1000 drops the bar to $6/1005 \approx 0.006$ — the direction that matters: as the miss-to-false-alarm ratio grows the threshold *falls*, driving the operator toward inspecting everything ambiguous (Figure 6); this direction follows directly from the cost inequality. *Now you try the other knob:* hold C_{miss}=100 and raise the false-alarm cost to C_{fa}=20 — does the bar rise or fall, and by how much? (It rises, to $21/120 = 0.175$: attention that is expensive to waste buys back the operator's right to pass.)

What this does not say. The theorem does not say the operator should inspect more. It

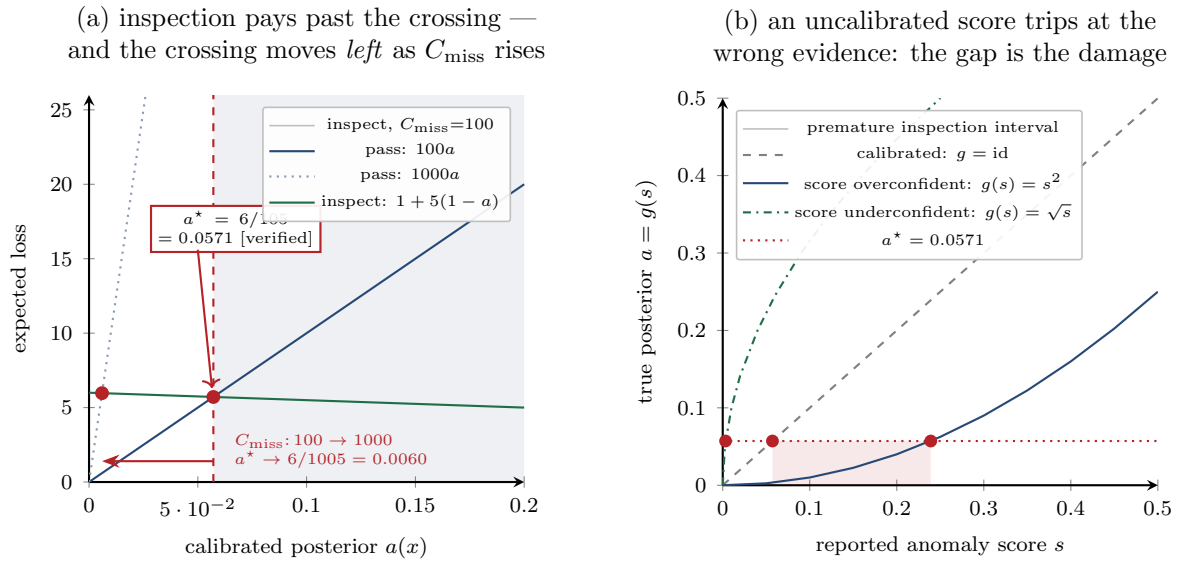


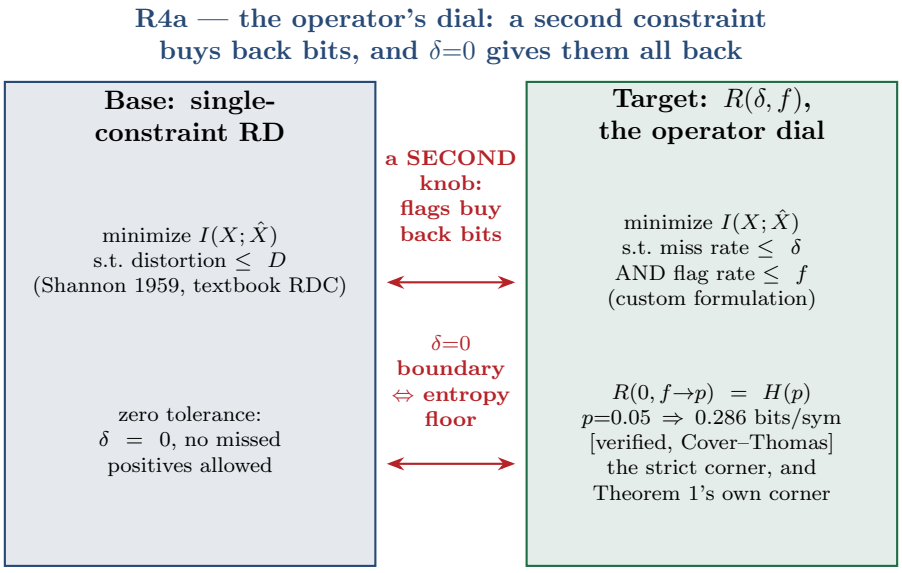
Figure 6: Regime diagram for the surfacing rule. Panel (a): expected loss of pass against expected loss of inspect, as a function of the calibrated posterior $a(x)$. Inspection pays exactly where the two loss lines cross, at $a^\star = (c_{\text{att}} + C_{\text{fa}})/(C_{\text{miss}} + C_{\text{fa}})$ — the closed form obtained by solving Theorem 3's inequality directly. At the shown costs (C_{miss}=100, c_{att}=1, C_{fa}=5) that is $6/105 \approx 0.057$; raising C_{miss} to 1000 moves the crossing *left*, to $6/1005 \approx 0.006$ [verified] — the counterintuitive direction, drawn rather than asserted. Panel (b): threshold the reported score s only after calibrating the true posterior $g(s)$. When $g(s) = s^2$, the naive rule $s \geq 0.057$ inspects too early: the true posterior is only about 0.00327 there; the correct score threshold is $s \geq \sqrt{0.05714} \approx 0.239$. The shaded interval $[0.057, 0.239)$ is unnecessary inspection. For $g(s) = \sqrt{s}$, the correct score threshold is about 0.00327 and the naive rule acts too late.

says the *threshold* moves; whether a moving threshold means more opens depends entirely on where the posterior mass sits, and an uncalibrated a moves the threshold without moving the decision it was supposed to make. Nor does it license tuning: with a miscalibrated anomaly score the product form is wrong by precisely the miscalibration, so fitting g from the audit log is a measured obligation and not a footnote (§7).

Claim 2 (Reputation enters only through the posterior)**.** *If agent reputation r is admitted, it enters as $a(x) = \Pr(\mathsf{bad} \mid \textit{evidence}, r)$ — never as a free multiplier on the score. High reputation lowers surfacing frequency only where the calibrated posterior actually falls, and cannot zero the passing-loss term when $a(x) > 0$; inspection still requires the cost inequality above. This is the formal content of "reputation must not defeat hard stakes gates."*

5 The digest–zoom frontier, and the zoom theorem

Theorem 1 prices one corner: zero misses at a fixed open budget. The operator's real dial has three settings — digest bits, expected opens, tolerated misses — and the exchange rates among them are this section's subject. Its second half proves a constant-explicit bound for adaptive zoom.



Two faces of one dial: the constraint that got added, and the boundary where it collapses back to Shannon. $R(0, 0.10) = 0.186$, $R(0.01, 0.10) = 0.110$, $R(0.04, 0.06) = 0.009$ *bits/sym [internal,* `b1_frontier.py`*].*

Figure 7: The relation-map for the two-constraint program. The relation that transfers from classical rate–distortion is not the objective — that is unchanged, $I(X;\hat{X})$ — but the shape of the feasible set: pricing false negatives separately from flag rate adds a second binding constraint, and every bit the digest saves is bought by loosening one of the two. Row 2 is the seam with Theorem 1: at $\delta{=}0$ and $f \to p$ the program returns the source's entire entropy, which is the frontier's own version of the covering floor.

5.1 The two-constraint program and its pinned closed form

Model each artifact's critical status as i.i.d. $X \sim \text{Bern}(p)$; a digest emits a per-item flag $\hat{X} \in \{0, 1\}$. Price the two failure modes separately:

$$R(\delta, f) \;=\; \min_{P_{\hat{X}|X}} \; I(X;\hat{X}) \quad \text{s.t.} \quad \Pr(X{=}1, \hat{X}{=}0) \le \delta, \qquad \Pr(\hat{X}{=}1) \le f.$$

This is a *custom formulation*, deliberately: classical rate–distortion charges one scalar distortion, which cannot express that oversight prices a missed critical differently from a wasted open. An August-2026 literature sweep found no prior false-negative-constrained lossy compression for binary sources under this framing; the nearest neighbors are expansion coding with one-sided error for continuous sources [6] and the rate–distortion–classification line for Bernoulli sources [7]. We position $R(\delta, f)$ against that literature as a formulation, not as a textbook import — and "not found" is not "proven nonexistent." What the second constraint changes, and where it collapses back to Shannon, is Figure 7.

Proposition (pinned joint). For $X \sim \text{Bern}(p)$ with $0 \le \delta < p$ and $p - \delta \le f < 1 - \delta$, if f is below the flag rate at which an X-independent flagger meets the miss constraint (i.e. $f < 1 - \delta/p$), both constraints bind at the optimum and the minimizing joint is fully pinned:

$$q_{11} = p - \delta, \qquad q_{10} = \delta, \qquad q_{01} = f - (p - \delta), \qquad q_{00} = 1 - f - \delta,$$

feasible iff all four are nonnegative, and $R(\delta, f) = I(X;\hat{X})$ evaluated at this joint. At the zero-miss tightest-flag corner, $R(0, f{\to}p) = H(p)$: the digest must spend the source's full

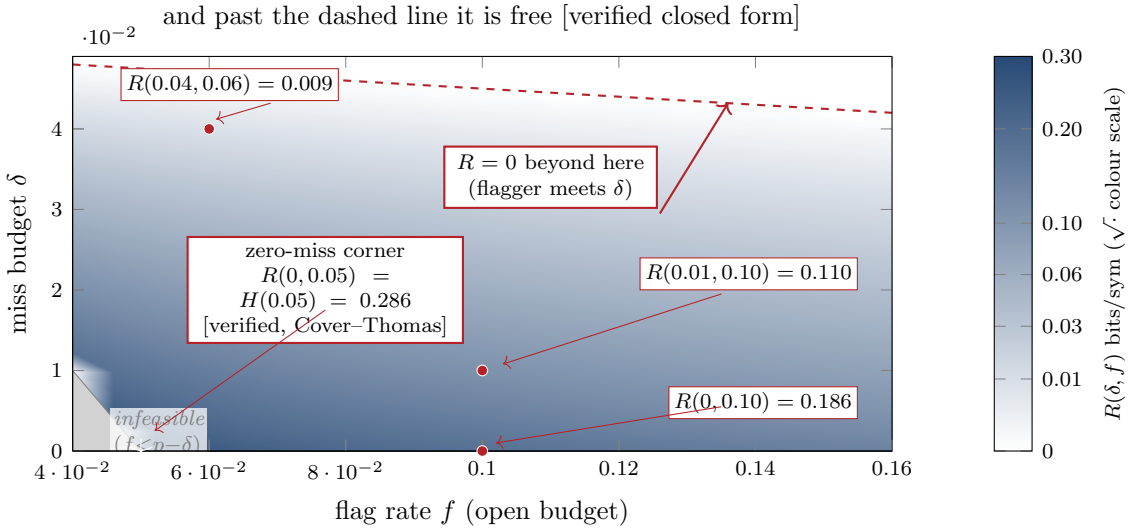


Figure 8: The two-constraint frontier $R(\delta, f)$ over the whole (f, δ) plane near the guarantee corner, from the Proposition's closed form (same formula as `b1_frontier.py`). The four numbers of the "Numbers" paragraph are marked directly on the surface, including the zero-miss corner that ties this section back to Theorem 1. The dashed line is the Proposition's own boundary $f = 1 - \delta/p$: below it both constraints bind and the pinned joint applies; above it an X-independent flagger already meets the miss budget and $R = 0$ exactly, not rising — the region the corrected geometric-honesty note above refers to. The small grey wedge is simply infeasible ($f < p - \delta$: not enough flag budget to cover the source rate at all).

entropy.

Proof. $I(X; \hat{X})$ is convex in $P_{\hat{X}|X}$ and decreases along the feasible ray toward the independent coupling ($I = 0$ iff independent). An independent flagger $\hat{X} \sim \text{Bern}(\phi)$ has miss mass $p(1 - \phi)$, so it meets the constraint only when $\phi \geq 1 - \delta/p$; when $f < 1 - \delta/p$ that coupling is outside the flag budget, so the minimum sits on the boundary where any remaining slack in either constraint could otherwise be spent moving further along the ray — both constraints bind. Binding constraints fix $q_{10} = \delta$ and $q_{11} + q_{01} = f$; with the source margin $q_{10} + q_{11} = p$ this pins all four cells. At $(\delta, f) = (0, p)$ the joint is diagonal, $\hat{X} = X$, and $I = H(p)$. □

Numbers ($p = 0.05$): $R(0, 0.05) = H(0.05) = 0.286$ bits/symbol [verified, Cover–Thomas [5]]; $R(0, 0.10) = 0.186$; $R(0.01, 0.10) = 0.110$; $R(0.04, 0.06) = 0.009$ [internal, `b1_frontier.py`]. Reading: loosening the open budget cheapens the digest; tolerating misses cheapens it dramatically — two orders of magnitude between the guarantee corner and $(\delta, f) = (0.04, 0.06)$ (Figure 8 shows the whole surface these four numbers sit on). Consistency with Theorem 1: at N=1000, $R \cdot N \approx 186$ bits at $(0, 2p)$, below the worst-case covering floor $\log_2 \binom{1000}{50} \approx 282$ bits — average-case rate under the worst-case floor, as it must be. The pinned closed form holds only while $f < 1 - \delta/p$, the Proposition's stated regime where both constraints bind; past that line an X-independent flagger already meets the miss budget on its own and the true rate is exactly 0, not rising. Applying the pinned formula beyond this domain would create a spurious uptick near $\delta/p \to 1$ instead of the true zero-rate region.

5.2 The zoom-advantage theorem

A conservative digest over-flags to be safe: it emits a flagged set of size F containing the k true criticals with $k \ll F$. The naive second stage opens all F. The adaptive second stage plays twenty questions with group queries (Figure 9): open a *group* of flagged items at once — an aggregate check cheap at group level, such as a combined diff scan or batched test run — learn whether the group contains any critical, and split only the groups that do. The analogy transfers because its critical relation transfers: halving a set costs one question and eliminates half the candidates *only when positives are rare in the set*. The misread to preempt: twenty questions finds *one* answer in $\log_2 F$ queries; with k answers the tree forks, and the accounting below is precisely the price of the forking.

Theorem 4 (zoom advantage; new). Let a flagged set of F items contain an unknown set P of $k \geq 1$ positives, and let a group query on S report whether $S \cap P = \emptyset$. Adaptive halving — query the flagged set; discard any group that tests negative; report any positive singleton; otherwise split the group into halves of sizes $\lfloor |S|/2 \rfloor$, $\lceil |S|/2 \rceil$ and recurse on both — identifies P exactly, using a total number of group queries

$$Q \;\leq\; 2k\,\lceil \log_2(F/k) \rceil + 4k \qquad (\text{and always } Q \leq 2F - 1),$$

for every placement of the k positives, versus F opens for flat inspection. The leading constant is tight: for F and k powers of two, evenly spread positives force $Q = 2k \log_2(F/k) + 2k - 1$.

Proof. Every group the algorithm ever queries is a node of the *halving tree* T of $[F]$: the root is the full flagged set, each internal node's children are its two halves, leaves are singletons. Since a node of size s has children of size at most $\lceil s/2 \rceil$, every node at depth d has size at most $\lceil F/2^d \rceil$, so T has depth $D \leq \lceil \log_2 F \rceil$, and T has exactly $2F - 1$ nodes (a binary tree with F leaves), giving the unconditional bound.

Call a node *positive* if its group meets P. The algorithm splits exactly the positive non-singleton nodes it visits, and every queried node is either the root or a child of a split node. Hence, writing S^+ for the number of positive non-singleton nodes of T,

$$Q \;\leq\; 1 + 2S^+.$$

It remains to bound S^+ by a charging argument. The nodes at any fixed depth d of T are pairwise disjoint subsets of $[F]$, so each positive item lies in at most one of them; therefore at most $\min(k, 2^d)$ nodes at depth d are positive. Charge the positive nodes in two regimes:

- *Top of the tree* ($d \leq \lfloor \log_2 k \rfloor$): charge each positive node to its level. The total over these levels is at most $\sum_{d=0}^{\lfloor \log_2 k \rfloor} 2^d \leq 2k - 1$.
- *Below the fan-out* ($d > \lfloor \log_2 k \rfloor$): charge each positive node to one positive item it contains. By disjointness within a level, each of the k positives absorbs at most one charge per level. Non-singleton nodes exist only at depths $d \leq D - 1$, so the number of charging levels is

$$D - 1 - \lfloor \log_2 k \rfloor \;<\; (\log_2 F + 1) - 1 - (\log_2 k - 1) \;=\; \log_2(F/k) + 1$$

(using $D \leq \lceil \log_2 F \rceil \leq \log_2 F + 1$ and the strict $\lfloor x \rfloor > x - 1$); an integer strictly below $\log_2(F/k)+1$ is at most $\lceil \log_2(F/k) \rceil$, so these levels contribute at most $k\lceil \log_2(F/k) \rceil$ charges.

Summing, $S^+ \leq k\lceil \log_2(F/k) \rceil + 2k - 1$, so $Q \leq 1 + 2S^+ \leq 2k\lceil \log_2(F/k) \rceil + 4k - 1$, which gives the bound. Correctness is immediate: a discarded group is certified free of positives, and every positive survives discarding at every level, so each is eventually isolated and reported as a positive singleton.

For tightness, take $F = 2^a$, $k = 2^b$, $b \le a$, positives at positions $\{iF/k : 0 \le i < k\}$. Every node at depth $d \le b$ contains a positive (2^{b-d} of them), so all $2^{b+1} - 1 = 2k - 1$ nodes down to depth b are queried and split (or, at depth b, split further); below depth b, exactly k nodes per level are positive, each is split at levels $b, \dots, a-1$, and each split queries two children. Counting queried nodes: all $2k - 1$ nodes at depths $0..b$, plus $2k$ children per level for levels $b{+}1, \dots, a$, of which per level k are positive; total queried $= (2k - 1) + 2k(a - b) = 2k \log_2(F/k) + 2k - 1$. This configuration is realized numerically below. □

Corollary 1 (The advantage, priced). Flat inspection opens F; adaptive halving opens at most $2k\lceil \log_2(F/k) \rceil + 4k$, so the guaranteed advantage is at least $F/(2k\lceil \log_2(F/k) \rceil + 4k)$ — at $(F, k) = (2500, 10)$, at least $2500/200 = 12.5\times$, against a measured mean of $15.3\times$ [internal, `b1_frontier.py`]. Written in the flagged-set density $d = k/F$ the advantage depends on nothing else:

$$\mathrm{adv}(d) \;=\; \frac{1}{d\big(2\lceil \log_2(1/d) \rceil + 4\big)},$$

which exceeds 1 exactly on $d < 1/12$ and equals 1 at $d = 1/12$ (the step function $d(2\lceil \log_2(1/d) \rceil + 4)$ first touches 1.0000 at $d = 1/12$) [verified]. The worked point above is d=0.004 — a far sparser flagged set than §5.1's p=0.05 frontier can deliver, which is the composition window priced below.

Positioning against group testing. None of the group-testing machinery here is new mathematics: Dorfman introduced pooled testing in 1943 [1], and Hwang's generalized binary splitting [2] finds k defectives in $\log_2 \binom{F}{k} + O(k)$ tests — essentially matching the counting lower bound $\lceil \log_2 \binom{F}{k} \rceil \ge k \log_2(F/k)$ that any adaptive strategy with binary outcomes must pay, with the full theory in Du and Hwang [3]. Plain halving pays a factor $2 + o(1)$ over Hwang because it re-queries both children of every split rather than inferring one subgroup and binary-searching within the other. Theorem 4's contribution is therefore *not* a better group-testing algorithm; it is the explicit worst-case constant for the exact procedure the digest architecture runs, proved tight — which converts the simulation's observed $\approx 2\times$ gap over the ideal $k \log_2(F/k)$ from an empirical curiosity into the algorithm's provable constant, and makes the frontier's second-stage cost a theorem rather than a fit.

Numbers, verified. A dedicated check (seed 20260816) re-runs the exact halving procedure of `b1_frontier.py`: at F=2500, k=10, mean 162.6 opens over 150 random placements ($15.4\times$ advantage, reproducing the compendium's $15.3\times$; ideal $k \log_2(F/k) = 79.7$; empirical overhead $2.04\times$), maximum observed 183, all under the bound $2 \cdot 10 \cdot 8 + 40 = 200$; random sweeps over $F \in [16, 5000]$, $k \in [1, 50]$ and adversarial evenly-spread and clustered placements never exceed the bound; and the tightness configuration F=4096, k=32 lands exactly on $2k \log_2(F/k) + 2k - 1 = 511$ [internal, `zoom_bound_check.py`]. Now you try the $k = 1$ sanity case by hand: $F = 1024$, one positive — halving queries two children per level for ten levels plus the root, 21 opens, under the bound $2 \cdot 10 + 4 = 24$, versus 1024 flat.

What each stage buys, priced separately. Together, §5.1 and Theorem 4 give the operator a three-way dial. Stage one, the digest, spends $R(\delta, f) \cdot N$ bits to cut N artifacts to $F \approx fN$ flags at miss budget δ; stage two, zoom, spends at most $2k\lceil \log_2(F/k) \rceil + 4k$ group opens to isolate the k criticals among the flags. Each stage's own price is exact: loosening flags from f=p to f=$2p$ cuts the digest from 0.286 bits/symbol [verified] to 0.186 [internal] at zero miss, and tolerating δ=0.01 misses cuts the rate almost in half again. The stages also close the loop with Theorem 1: the worst-case covering floor is the frontier's (δ=0, $f \to p$) corner.

The composition window, stated. The two prices are drawn from one budget, and they compose only inside a window this paper can state exactly. Stage one sets stage two's density: by the Proposition both constraints bind, so the flagged set holds $(p - \delta)N$ real criticals among

fN flags, and

$$d = \frac{k}{F} = \frac{p-\delta}{f}, \qquad \text{guaranteed advantage} = \text{adv}(d) = \frac{1}{d\big(2\lceil \log_2(1/d)\rceil + 4\big)}.$$

Since $\text{adv}(d) > 1$ exactly on $d < 1/12$, *the bound guarantees a strict saving if and only if the flag budget exceeds twelve times the residual positive rate*, $f > 12(p-\delta)$ [verified from the two closed forms]; the measured mean crossover is about twice as forgiving, $d \approx 0.165$, i.e. $f \gtrsim 6(p-\delta)$ [internal, `zoom_bound_check.py`]. At p=0.05, δ=0 the bound is break-even at $f = 0.60$ — the digest must flag more than 60% of the night's artifacts before the bound guarantees beating flat inspection. §5.1's recommended loosening does not reach it: at f=$2p$ the density is 0.500 and the guarantee is 0.33× (a 3× loss, not a saving); at $(\delta, f) = (0.01, 0.10)$ the density is 0.400 and the guarantee is 0.31×. *Inside* the window the flags stage one adds are cheap in the second stage — k is fixed by the source, so doubling F costs only the logarithmic term, $2k$ further opens. *Outside* it, the honest second-stage cost is linear in F, because reading all F flags is then the better procedure. The Corollary's 12.5× belongs to d=0.004, two-and-a-half orders of magnitude sparser than anything the p=0.05 frontier produces: a source with $p \approx 0.004$ and a generous flag budget. Stated plainly: cheap digests and cheap zooms are bought from the same budget, and the operating points where both are cheap are exactly the sparse-source ones.

The dense-flag boundary, stated. At the other end the advantage inverts outright. When the flagged set is already dense in positives (k/F large — e.g. a flag budget barely above the residual positive rate, $f \approx 1.11(p-\delta)$, so nine flags in ten are real), group overhead dominates and adaptive halving opens *more* than F: at F=100, k=90, density d=0.9, the run costs 199 opens against 100 flat [internal] — exactly Theorem 4's unconditional $2F-1$, reached because at that density essentially every node of the halving tree is positive. The regime of strict worst-case savings from this bound is the sparse-flagged corner $d < 1/12$ (Figure 10). A miss-averse digest moves toward that corner by over-flagging, but enters the strict-guarantee region only if it over-flags by more than a factor of twelve; the modest loosening §5.1 prices does not get there. The theorem supports the aggressively over-flagging operating point, and honestly does not support the others.

6 Related work

Group testing. Dorfman [1], Hwang's generalized binary splitting [2], and the Du–Hwang monograph [3] supply the second stage's mathematical substrate; §5.2 states precisely what is imported (the algorithmic idea and the counting lower bound) and what is contributed (the constant-explicit worst-case bound for plain halving, its tightness, and its role as the frontier's second-stage cost). **Rate–distortion.** The single-constraint theory is classical [4, 5]. The two-constraint $R(\delta, f)$ is a custom formulation for oversight; the nearest published lines are expansion coding with one-sided error for continuous sources [6] and rate–distortion–classification theory for Bernoulli sources [7], neither of which prices false negatives as a separate constraint for binary sources. We claim the formulation and its pinned closed form as this paper's framing contribution, subject to the standard caveat that a survey's "not found" is defeasible. **Attention and oversight.** That attention is the scarce resource an information-rich environment consumes is Simon's observation [8]; competitive analysis of paging [9] inspires the budgeted-opens framing. On the AI-oversight side, debate and related protocols [10] and delegated-computation results [11] ask how a weak verifier can check strong provers; this paper is complementary — it prices the *interface* (the digest) rather than the protocol, and its floors apply to any oversight scheme whose human reads a bounded summary. **Signal detection.** Theorem 3 is a Bayes-risk likelihood-ratio test, textbook decision theory; the contribution is the exact identification of the folklore product form with it and the resulting calibration obligation, with the direction of the threshold fixed by the cost inequality. The theorem is single-shot (Neyman–Pearson); oversight is a repeated-observation stream, and the natural sequential extension, with time-varying

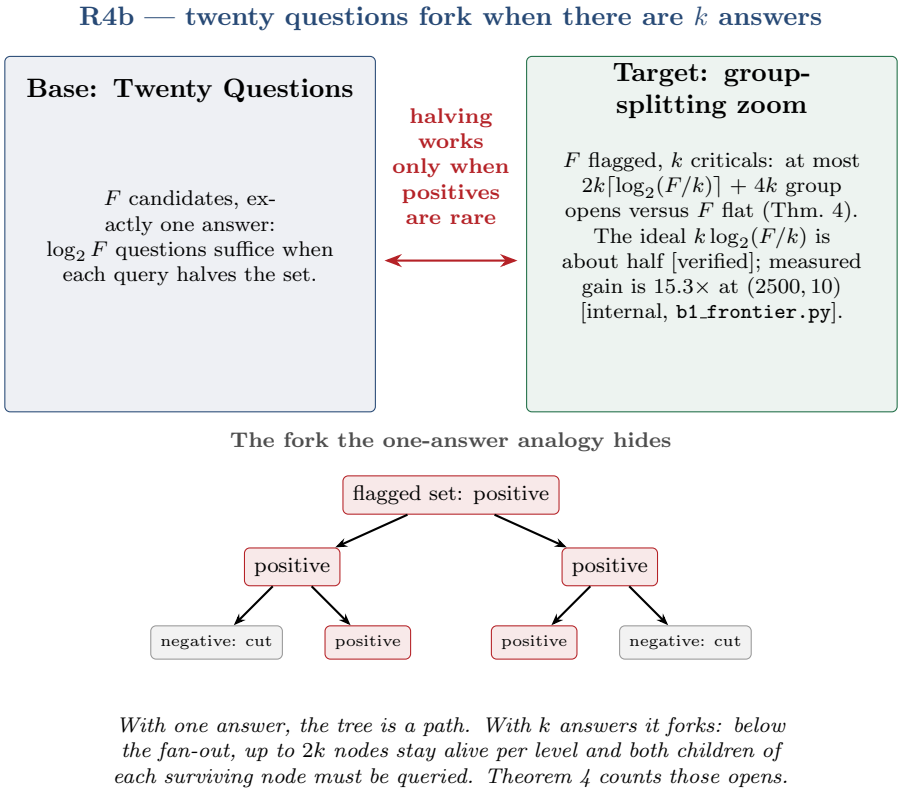


Figure 9: The relation-map for zoom: twenty questions (base) against group-split inspection of the flagged set (target). The correspondence that does the work is "one question halves the candidate set *when answers are rare*"; the misread it invites — that $\log_2 F$ questions suffice — is exactly what Theorem 4's $2k\lceil\log_2(F/k)\rceil + 4k$ accounting corrects for $k > 1$, and the drawn fork is where the extra factor comes from. Note the target cell states the *deployed* bound: the ideal $k\log_2(F/k)$ is a factor ≈ 2 cheaper and is not what the digest architecture runs.

thresholds and optimal expected sample size under fixed error rates, is Wald's Sequential Probability Ratio Test [14] — not developed here. **Comonotonicity.** Theorem 2's proof is derived from scratch, but the property it names — two preference orders that never cross — is decision theory's *comonotonicity*, defining rank-dependent and Choquet-expected utility since Schmeidler and Yaari [12, 13]; the super-additive pricing of Claim 1 is, to our knowledge, new against that thirty-five-year-old literature.

7 Limitations and honest boundaries

Honest boundary — stated at the same prominence as the claims.

- *Theorem 1* lower-bounds the *zero-miss guarantee* in a worst-case combinatorial model with a data-independent decoder; it says nothing against cheaper digests at nonzero miss (that is §5.1's job) and nothing average-case.
- *Theorem 2* forbids a shared head only for non-comonotone readers; comonotone readers can share. The worked extractive instance establishes that such a crossing is possible; prevalence in deployed fleets is an empirical question.
- *Theorem 3* is exactly as good as its calibration: with an uncalibrated anomaly score the product form is wrong by precisely the miscalibration, and fitting g from the audit log is a measured obligation, not a footnote.

R4 regime — the two-constraint dial and the boundary where zooming stops paying

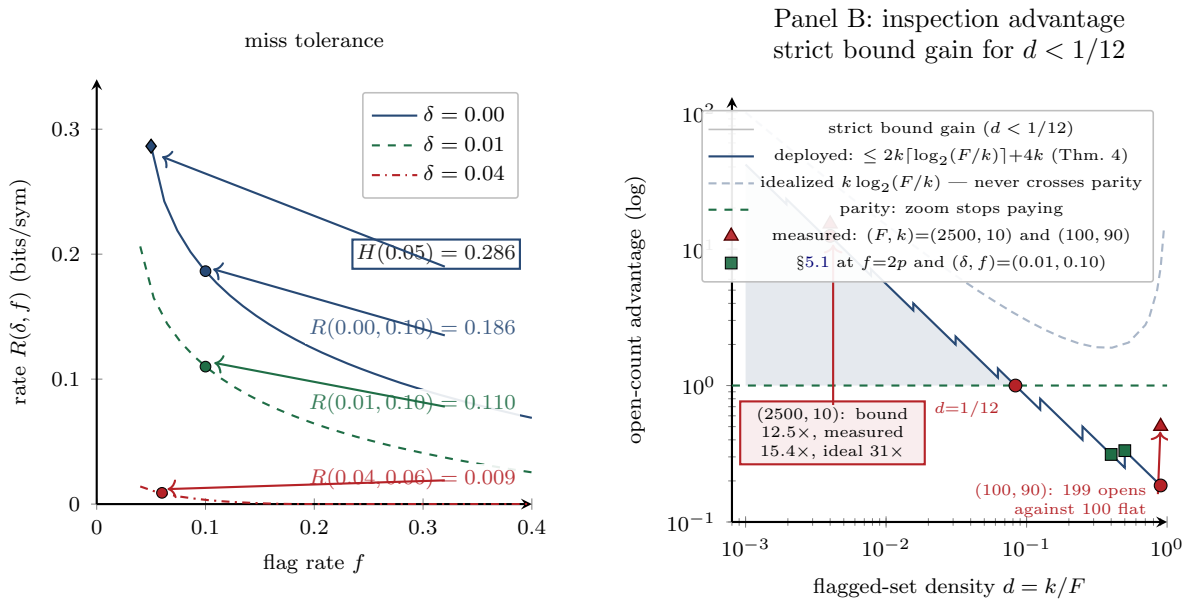


Figure 10: Panel A shows the Bernoulli rate–distortion frontier with separate miss and flag budgets, including its zero-rate region. Panel B plots the guaranteed advantage $1/[d(2\lceil\log_2(1/d)\rceil + 4)]$ of the specified halving procedure, with the idealized $1/[d\log_2(1/d)]$ for comparison. The displayed bound guarantees a strict saving for $d < 1/12$ and breaks even at $d = 1/12$. At dense placement $(F, k) = (100, 90)$ the measured procedure uses 199 opens against 100 flat; a bound below parity alone would not prove that loss.

- *The frontier* $R(\delta, f)$ is an i.i.d. Bernoulli model of critical status — a modeling commitment, not a law; and the formulation is custom, positioned against [6, 7], not a textbook theorem.
- *Theorem 4* assumes a reliable group query at unit cost; correlated or noisy group checks, and group costs growing with $|S|$, change the accounting. The advantage inverts in the dense-flag regime, as stated and measured, and it composes with §5.1's frontier with a strict guarantee only for $f > 12(p - \delta)$ — at p=0.05 that is $f > 0.60$, well outside the flag budgets §5.1 tabulates, so the two halves of §5 are not jointly instantiated at the same source rate anywhere in this paper. Plain halving is a factor ≈ 2 from Hwang's optimum by design — we bound the deployed algorithm, not the best one.
- *Falsification scope:* the 0/16 result tests the floor's encoder/decoder formalization at one swept regime (N=60, k=2, m=8); the noisy-encoder curve in that experiment barely beats random at SNR ≈ 1 and must not be read as evidence noisy digests are near-optimal. The feature-corruption panel of the same experiment is visually compressed at m=8 (the budget is generous enough that corruption barely moves in-range miss) and would need a tighter-m rerun for a publication-grade figure.

8 Reproducibility

Internal results use seed `20260816`. They regenerate from `a7_experiment.py` (the falsification experiment; 0/16) and `b1_frontier.py` (the $R(\delta, f)$ table and the zoom simulation), both bundled with the research program's results skill, plus this paper's companion check `zoom_bound_check.py` (Theorem 4's bound against random and adversarial placements, the

exact tightness configuration, and the dense-flag boundary). Every [verified] number is recomputable from the stated closed forms with a hand calculator. Figures regenerate from `figures/src/` at the same seed.

References

[1] R. Dorfman. The detection of defective members of large populations. *Annals of Mathematical Statistics*, 14(4):436–440, 1943.

[2] F. K. Hwang. A method for detecting all defective members in a population by group testing. *Journal of the American Statistical Association*, 67(339):605–608, 1972.

[3] D.-Z. Du and F. K. Hwang. *Combinatorial Group Testing and Its Applications*, 2nd ed. World Scientific, 2000.

[4] C. E. Shannon. Coding theorems for a discrete source with a fidelity criterion. *IRE National Convention Record*, part 4, 142–163, 1959.

[5] T. M. Cover and J. A. Thomas. *Elements of Information Theory*, 2nd ed. Wiley, 2006.

[6] H. Si, O. O. Koyluoglu, and S. Vishwanath. Lossy compression of exponential and Laplacian sources using expansion coding. arXiv:1308.2338, 2013.

[7] Rate-distortion-classification representation theory for Bernoulli sources. arXiv:2601.11919, 2026; see also the universal rate-distortion-classification line, IEEE ITW 2025.

[8] H. A. Simon. Designing organizations for an information-rich world. In M. Greenberger, ed., *Computers, Communications, and the Public Interest*. Johns Hopkins Press, 1971.

[9] D. D. Sleator and R. E. Tarjan. Amortized efficiency of list update and paging rules. *Communications of the ACM*, 28(2):202–208, 1985.

[10] G. Irving, P. Christiano, and D. Amodei. AI safety via debate. arXiv:1805.00899, 2018.

[11] S. Goldwasser, Y. T. Kalai, and G. N. Rothblum. Delegating computation: interactive proofs for muggles. In *STOC 2008*, 113–122.

[12] D. Schmeidler. Subjective probability and expected utility without additivity. *Econometrica*, 57(3):571–587, 1989.

[13] M. E. Yaari. The dual theory of choice under risk. *Econometrica*, 55(1):95–115, 1987.

[14] A. Wald. *Sequential Analysis*. Wiley, 1947.